Locality at $n = 9$ from the cyclic 1-zeros

via the block-rule Laurent cascade

(satellite for the $\mathrm{Tr}(\phi^3)$ locality program)

Jonathan Valenzuela

Abstract

We prove locality at $n = 9$ from the cyclic 1-zero conditions alone: every non-local coefficient of the most general rational ansatz with mass dimension $-2(n-3)$ and at most simple planar poles vanishes once the 9 cyclic hidden-zero constraints are imposed. The proof combines Step-1 (Layer-0) elimination with a depth-1 Laurent cascade. Step-1 kills 904,752 of the 905,763 non-triangulation size-6 multisets directly. The remaining 1,011 multisets fall into 113 cyclic-orbit equivalence classes, each non-local. Of these, 23 orbits admit independent single-orbit depth-1 cascade recipes; the other 90 orbits form a coupled cluster $\mathcal{C}_{90}$ whose depth-1 fingerprint matrix has 506 rows and column rank 90 on cluster columns, forcing the cluster's 90 coefficients to vanish jointly. A complete per-orbit audit confirming all 113 kill mechanisms is recorded in [1]. *This is the first polygon size at which the cluster mechanism is required:* at $n = 5, 6$ Step-1 alone closes locality; at $n = 7, 8$ every Step-1 survivor admits a single-orbit cascade (cluster sizes 1); at $n = 9$ a single 90-orbit non-trivial cluster appears and is killed jointly by the depth-1 fingerprint system. Unitarity (equality of triangulation coefficients to a single common scalar) is independent of this paper and follows from the d-subset / homogeneity-rescaling argument of [6, 7, 4]; locality is the load-bearing half of the $\mathrm{Tr}(\phi^3)$ uniqueness theorem and is what this paper closes at $n = 9$.

1 Background: the three layers of kill mechanism

We adopt the conventions of the root README. The cyclic hidden-zero zone $\mathcal{Z}_{r,r+2}$ has special chord $X_{r,r+2} =: X_*$, bare chord $X_{r+1,r-1} = -X_*$, and substitutions $X_{r+1,k} = X_{r,k} - X_*$ for k in cyclic range. The general ansatz is $B = \sum_M a_M / \prod_{c \in M} X_c$ over size-$(n-3)$ chord multisets M.

Definition 1 (Three layers of kill mechanism)**.**

- **Step-**1 **(Layer-**0**)**: at zone $\mathcal{Z}_{r,r+2}$, send $X_* \to \infty$. The leading-order coefficient identity forces $a_M = 0$ when $M \in \mathcal{K}_{r,r+2}$, i.e. when (K1) M contains neither special nor bare and (K2) M contains no (companion, substitute) pair. Holds for M killable at *any* cyclic zone.
- **Step-**2 **(equate)**: keep X_* in the surviving variables and exploit $X_{r+1,r-1} = -X_*$. This produces equality identities $a_{M \cup \{X_{r+1,r-1}\}} = -a_{M \cup \{X_*\}}$ between coefficients differing by a bare/special swap. Step-2 is the unitarity engine; locality does *not* require it.
- **Step-**3 **(Laurent cascade)**: at zone $\mathcal{Z}_{r,r+2}$, expand

$$\frac{1}{X_{r+1,k}} = \frac{1}{X_{r,k} - X_*} = -\frac{1}{X_*} - \frac{X_{r,k}}{X_*^2} - \frac{X_{r,k}^2}{X_*^3} - \cdots$$

  and extract the coefficient of a free-variable monomial (the *fingerprint*) at order $X_*^{-(\ell_r(M)+1)}$. This is a linear identity among coefficients a_M of multisets contributing the same fingerprint.

Definition 2 (Cyclic orbit; cousin)**.** $\mathbb{Z}_n$ acts on size-$(n-3)$ multisets by vertex shift; an *orbit* is a $\mathbb{Z}_n$-equivalence class. Given a Step-3 fingerprint equation $\sum_C \alpha_C\, a_C = 0$ at $(\mathcal{Z}, k, \mathcal{F})$, the multisets $\{C : \alpha_C \neq 0\}$ are the *cousins* of any M in the equation; their orbit-classes are the *cousin orbits*.

Remark 3 (Step-1 exhaustivity at low n)**.** At $n = 4, 5, 6$ Step-1 alone kills every non-triangulation coefficient—the count of non-triangulation Step-1 survivors is 0 at each of these polygon sizes [2]. Cluster and cascade machinery are required only starting at $n = 7$.

2 Single-orbit cascade and block-rule cascade

Definition 4 (Single-orbit cascade)**.** Orbit $\mathcal{O}$ admits a *single-orbit depth*-1 *cascade* recipe iff there is a triple $(\mathcal{Z}, k, \mathcal{F})$ with $k = \ell_{\mathcal{Z}}(M) + 1$ for some $M \in \mathcal{O}$ such that: (i) the fingerprint equation has a_M with nonzero scalar; and (ii) every cousin of M in this equation is Step-1-killable at some cyclic zone.

When (i)+(ii) hold, substituting cousin-zeros into the fingerprint equation collapses it to $\alpha_M\, a_M = 0$, so $a_M = 0$. Cyclic equivariance rotates the recipe onto every member of $\mathcal{O}$.

The mechanism that closes $n = 7, 8$ is: *every Step*-1 *survivor orbit admits a single-orbit recipe.* The $n = 9$ data falsifies this for 4 orbits, and a further 14 orbits time out under a 600-second search cap; the $4 + 14$ all lie in a single coupled cluster, and are killed jointly by the block-rule below rather than by individual cascades.

Definition 5 (Coupling graph; cluster)**.** The *coupling graph* G_n has vertex set = Step-1 survivor orbits at n. An edge $\mathcal{O} \sim \mathcal{O}'$ exists iff there is a Step-3 fingerprint equation with $\mathrm{orbit}(M) = \mathcal{O}$ and $\mathrm{orbit}(C) = \mathcal{O}'$ for some non-local cousin C. A *cluster* is a connected component of G_n. A *singleton cluster* is a connected component of size 1 (an isolated vertex, with no non-local cousins in any Step-3 fingerprint equation).

Definition 6 (Block-rule cascade)**.** Let $\mathcal{C} = \{\mathcal{O}_1, \ldots, \mathcal{O}_m\}$ be a cluster. Enumerate all admissible depth-1 fingerprint recipes for orbits in $\mathcal{C}$ and assemble a matrix $M_{\mathrm{FP}}^{\mathcal{C}}$ with rows = fingerprint equations and columns = cluster orbits (*cluster columns*) plus external orbits (Step-1 survivors not in $\mathcal{C}$, plus triangulation orbits appearing as cousins). The cluster is *block-rule killable at depth*-1 iff the cluster sub-matrix has full column rank $m = |\mathcal{C}|$, given that each non-cluster non-local survivor column is set to 0 via its own single-orbit cascade.

The single-orbit cascade is the special case of the block-rule on a singleton cluster. Through $n = 8$ every cluster is a singleton; at $n = 9$ a non-trivial 90-orbit cluster appears alongside 23 singletons.

Proposition 7 ($n = 7$ baseline: singleton clusters)**.** *At $n = 7$, the seven Step*-1 *survivors form a single $\mathbb{Z}_7$ orbit. As a connected component of G_7 it is a singleton cluster: every cousin of the orbit's fingerprint equation is Step*-1*-killable. Equivalently, $\mathcal{C} = \mathcal{O}$ has $|\mathcal{C}| = 1$ and the 1×1 block-rule matrix has rank* 1*, forcing $a_{\mathcal{O}} = 0$.*

The proof is in `paper/step2 Laurent series for hard kills/cascade_n7.tex`. The same singleton structure (clusters of size 1) closes $n = 8$: all 13 cyclic orbits are singleton clusters, all killed by their own single-orbit cascades.

3 The $n = 9$ result

Step-1 enumeration at $n = 9$ yields 1,011 non-triangulation survivors organised into 113 cyclic orbits (one of size 3, $\mathbb{Z}_3$-stabilised; 112 of size 9, free orbits). Every representative is a non-local size-6 multiset: 98 are crossing configurations and 15 are double-pole configurations [1].

3.1 Single-orbit pass

The script `computations/step3_laurent/cascade_n9/cascade_kill_n9.py` attempts a single-orbit depth-1 recipe for each of the 113 representatives, with a 600-second timeout per orbit, then a diagnostic re-run with extended budget on the timeouts. The final disposition:

Outcome	count
single-orbit recipe found (after re-run)	95
search timeout (recipe still undetermined)	14
search exhausted, no recipe found	4 $\{\mathcal{O}_{22}, \mathcal{O}_{46}, \mathcal{O}_{88}, \mathcal{O}_{108}\}$

The 4 exhausted-search failures are diagnosed in `.../n9/diagnostics/diagnose_orbit_22.md`; they are not artifacts of bounded search depth or pruning. For example, $\mathcal{O}_{22}$ has representative $M_{22} = \{X_{1,3}, X_{1,4}, X_{1,7}, X_{1,8}, X_{4,6}, X_{5,7}\}$ and at every depth $1 \leq k \leq 4$ each fingerprint equation contains at least one cousin that is itself a Step-1 survivor; specifically $\mathcal{O}_{22}$ couples directly to $\mathcal{O}_{88}, \mathcal{O}_{96}, \mathcal{O}_{102}$ at the $\mathcal{Z}_{3,5}$ depth-1 fingerprint $X_{3,6}/(X_{1,3}X_{1,7}X_{1,8}X_{5,7})$. Single-orbit cascade cannot kill $\mathcal{O}_{22}$ in isolation; the four failures require collective treatment.

3.2 Cluster BFS and matrix

A breadth-first traversal of the coupling graph G_9 starting from $\mathcal{O}_{22}$ absorbs 90 orbits (all 4 failures and all 14 timeouts among them). The remaining 23 orbits are isolated nodes of G_9—singleton clusters—each killed by its own verified single-orbit recipe.

For the 90-orbit cluster $\mathcal{C}_{90}$, the depth-1 cascade matrix tabulates 506 fingerprint equations against 143 columns (90 cluster orbits + 32 triangulation orbits appearing as cousins + 21 non-cluster non-local survivor orbits). Working over $\mathbb{Q}$:

quantity	value
cluster size $\lvert\mathcal{C}_{90}\rvert$	90
depth-1 fingerprint equations	506
external columns (cluster-touched triangulation + non-cluster non-local)	$32 + 21 = 53$
$\mathrm{rank}(M_{\mathrm{FP}}^{\mathcal{C}_{90}})$ on cluster columns	**90**
cluster residual $r = \lvert\mathcal{C}_{90}\rvert - \mathrm{rank}$	**0**

Theorem 8 ($n = 9$ block-rule kill)**.** *The depth-1 fingerprint sub-matrix on the 90 cluster columns of $M_{\mathrm{FP}}^{\mathcal{C}_{90}}$ has full column rank 90. Consequently, given that the 21 non-cluster non-local survivor coefficients vanish (each by its own single-orbit cascade), every one of the 90 cluster non-local coefficients vanishes:*

$$a_{\mathcal{O}_i} = 0 \qquad \forall\, \mathcal{O}_i \in \mathcal{C}_{90}.$$

Proof. The rank computation is symbolic over $\mathbb{Q}$ via `sympy`; the trace is in `cluster_partial.json`. With non-cluster non-local survivor columns set to 0, the cluster matrix gives a homogeneous linear system on the 90 cluster columns plus 32 triangulation columns. Full column rank on cluster columns means: the assignment "triangulations equal a common scalar c, non-cluster non-locals $= 0$, cluster $= 0$" satisfies the system (because A_n^{tree} does, putting this vector in the nullspace), and this is the unique cluster-coefficient solution under any consistent triangulation assignment. □

3.3 Locality at $n = 9$, unconditional

Theorem 9 ($n = 9$ locality)**.** *Suppose $B\big|_{\mathcal{Z}_r} = 0$ for all nine cyclic zones at $n = 9$. Then $a_M = 0$ for every non-triangulation multiset M of size 6.*

Proof. Partition the 905,763 non-triangulation multisets at $n = 9$ by Step-1 killability:

(i) **Step-1 killed.** 904,752 multisets are Step-1 killable at some zone, so their coefficients vanish by Definition 1.

(ii) **Singleton-cluster Step-1 survivors.** The remaining 1,011 multisets form 113 cyclic orbits; 23 of these are singleton clusters (isolated nodes of G_9) and admit single-orbit depth-1 cascade recipes (§3.1, audited in [1]). Each recipe forces its orbit's coefficient to zero.

(iii) **Non-trivial cluster Step-1 survivors.** The remaining 90 orbits form $\mathcal{C}_{90}$. By Theorem 8, all 90 cluster non-local coefficients vanish.

Cases (i)–(iii) exhaust all non-triangulation multisets. □

Corollary 10 ($n = 9$ amplitude reconstruction)**.** *Combined with the d-subset / homogeneity-rescaling unitarity argument of [6, 7, 4], which equates all triangulation coefficients to a common scalar c, $B = c \cdot A_9^{\text{tree}}$.*

The unitarity input is logically independent of this paper. Locality (Theorem 9) closes the load-bearing half of the $\mathrm{Tr}(\phi^3)$ uniqueness claim at $n = 9$; the d-subset argument closes the unitarity half.

3.4 Comparison with $n = 7, 8$

	$n = 7$	$n = 8$	$n = 9$
Step-1 non-tri survivors	7	100	1,011
cyclic orbits	1	13	113
singleton clusters $\lvert\mathcal{C}\rvert = 1$	1	13	23
non-trivial clusters $\lvert\mathcal{C}\rvert > 1$	0	0	1 (size 90)
fingerprint equations on largest cluster	1	1	506
matrix rank on cluster columns	1	1	90
cluster residual r	0	0	0

The growth from $n = 7, 8$ to $n = 9$ is qualitative: at $n \leq 8$ the cluster mechanism is trivial ($|\mathcal{C}| = 1$ everywhere) and the locality proof reduces to the singleton cascade. At $n = 9$ a single non-trivial cluster of 90 orbits appears, and is killed jointly by the depth-1 system. The 5× ratio of fingerprint

equations to cluster orbits (506/90) overdetermines the cluster and gives residual $r = 0$. Whether this overdetermination persists at higher n, or whether a non-trivial residual ever appears, is the central all-n open question (§6).

4 Geometric interpretation

The cascade mechanism is the algebraic shadow of a geometric operation. At zone $\mathcal{Z}_{r,r+2}$, the substitution $X_{r+1,k} = X_{r,k} - X_*$ encodes the *ear-collapse* that identifies vertex $r+1$ with the rest of the polygon. The bare relation $X_{r+1,r-1} = -X_*$ is the 4-*gon kinematic identity*: the only chord relation at $n = 4$ is $X_{1,3} = -X_{2,4}$, and every higher-n collapse uses this identity at one local cyclic position.

The Laurent expansion in $1/X_*$ records the *kinematic memory* of the collapse: each non-bare substitute $X_{r+1,k}$ remembers its companion $X_{r,k}$ at order $X_*^{-(j+1)}$ (coefficient $X_{r,k}^j$). Depth-d probes the d-th layer of this memory.

A non-local multiset killable by a depth-d cascade has d chord-pairs whose crossing structure interacts with the relevant ear-collapse. A multiset *not* killable by any single-zone cascade—the cluster case—has crossings coupled across multiple cyclic orbits: configurations reachable from each other by a single ear-collapse move. The cluster is the connected component of the associahedron *face graph* restricted to ear-collapse moves carried by the chosen non-local cousins.

The fact that the depth-1 system on a cluster has rank $\geq |\mathcal{C}| - 1$ is, geometrically, the statement that a connected face of the associahedron has only finitely many "free directions" left after the cyclic-1-zero constraints are imposed. At small n the face is small enough that a 1-form might be needed to close the last degree of freedom (a possibility not yet observed in the data through $n = 9$); at $n = 9$ the 90-orbit face is rigidified by sheer constraint count.

5 Refined all-n locality conjecture

The $n = 9$ data sharpens the all-n statement.

Conjecture 11 (Block-rule cascade locality)**.** *For every $n \geq 7$ and every cluster $\mathcal{C}$ of Step-1 survivor orbits at n:*

(1) *the cluster $\mathcal{C}$ is finite;*

(2) *there exists $r = r(\mathcal{C}) \geq 0$ with $\operatorname{rank}(M_{\mathrm{FP}}^{\mathcal{C}}) = |\mathcal{C}| - r$ on cluster columns;*

(3) *if $r \geq 1$, the residual r-dimensional space is closed by r higher-Laurent-order anchor equations.*

Equivalently: every non-local coefficient of an n-point ansatz satisfying the n cyclic 1-zero conditions is forced to vanish by Step-1 together with the depth-1 block-rule cascade and a finite collection of higher-Laurent anchors.

Corollary 12 (All-n locality, conditional)**.** *Conjecture 11, together with the Step-1 exhaustivity recorded in [2] and the triangulation-survives-Step-1 lemma of [3], implies $a_M = 0$ for every non-triangulation M at every n.*

The conjecture is verified at $n = 4, 5, 6, 7, 8, 9$:

- $n = 4, 5, 6$: every non-local Step-1 killable; no Step-3 work needed.
- $n = 7, 8$: every cluster is a singleton ($|\mathcal{C}| = 1$), $r = 0$.
- $n = 9$: $|\mathcal{C}_{90}| = 90$, $r = 0$ (Theorem 8); plus 23 singleton clusters, each $r = 0$.

The pattern across these data points is consistent with the prediction $r = 0$ throughout: the fingerprint-equation count overdetermines clusters whenever they grow beyond singletons. A non-trivial residual ($r \geq 1$) has not been observed in any verified case and may not appear at any n.

6 What remains for the all-n locality theorem

The proof architecture is fully laid out. The remaining mathematical obligations for the all-n locality theorem are:

1. **Cluster finiteness.** Every cluster $\mathcal{C}$ at every n is finite. Likely follows from finiteness of size-($n-3$) chord multisets, but should be recorded as a lemma.
2. **Cluster rank lower bound.** Show $\operatorname{rank}(M_{\mathrm{FP}}^{\mathcal{C}}) \geq |\mathcal{C}| - 1$ on cluster columns for all n. This is the core combinatorial-geometric content. A generating-function argument counting fingerprint equations vs cluster orbits is plausible; an associahedron-face rigidity argument is the structural counterpart.
3. **Anchor existence when $r = 1$.** If a cluster ever has a 1-dimensional residual at any n, exhibit a higher-Laurent-order anchor that closes it. Through $n = 9$ no such case has been observed, but the all-n proof should accommodate the possibility.

These are the only obligations for locality at all n. Unitarity (equality of triangulation coefficients) is independent and is handled by the d-subset / homogeneity-rescaling argument; see [6, 7, 4].

Files and reproducibility

- `computations/step1_layer0_kill/kill_enumeration/` — Step-1 enumeration through $n = 9$.
- `computations/step3_laurent/cascade_n9/cascade_kill_n9.py` — the single-orbit cascade engine.
- `computations/step4_laurent_block_analysis/n9/scripts/` — the cluster-BFS, matrix-rank, and anchor-search scripts.
- `computations/step4_laurent_block_analysis/n9/outputs/` — `cluster_partial.json` (the rank-90 matrix trace), `cluster_partial.md` (human-readable summary), `results_cascade_n9_reps.json` (per-orbit cascade outcomes), `n9_locality_status.md` (**the consolidated per-orbit locality audit**).
- `computations/step4_laurent_block_analysis/n9/diagnostics/` — per-anomaly walkthroughs for orbits 22, 25, 28, 34, 35, 46, 88, 108.

References

[1] *Consolidated $n = 9$ locality status.* This repository, `computations/step4_laurent_block_analysis/n9/outputs/ n9_locality_status.md`.

[2] *Step-1 kill statistics and asymptotic locality.* Companion satellite, this repository, `paper/step1 Kill technique and statistics/step-one-kill-statistics/`.

[3] *Triangulations are not Step-1 killable.* Companion satellite, this repository, `paper/step1 Kill technique and statistics/step-one-doesnt-kill-triangulations/`.

[4] *Proof of Rodina's claim $B = 0 \Rightarrow B_i = 0$.* Companion note, this repository, `paper/Details missing in Hidden zero -> unitarity Rodina proof/Proof of Rodina claim B=0->`

[5] *The Laurent-tail kill at $n = 7$: seven layer-0 survivors and how they die.* Companion satellite, this repository, `paper/step2 Laurent series for hard kills/`.

[6] Handwritten note #11, *finale-d-subset-induction*, this repository, `notes/`.

[7] Handwritten note #18, *d-subset rigorous proof*, this repository, `notes/`.

[8] L. Rodina, *Hidden zeros are equivalent to enhanced ultraviolet scaling and lead to unique amplitudes in* $\mathrm{Tr}(\phi^3)$ *theory.* arXiv:2406.04234, 2024.

[9] N. Arkani-Hamed, Q. Cao, J. Dong, C. Figueiredo, S. He, *Hidden zeros for particle/string amplitudes and the unity of colored scalars, pions, and gluons.* arXiv:2312.16282, 2023.